

# Full Stack Development with MERN

## Project Documentation

### 1. Introduction

**Project Title:** Import Data using Transform Maps (Spreadsheet)

**Team Members:**

| Team Member     | Role                                            |
|-----------------|-------------------------------------------------|
| Yogesh P        | Team Lead – Data Import & Testing               |
| Tarun R         | Team Member – Data Validation & Documentation   |
| Ranjith R       | Team Member – Transform Map & Data Mapping      |
| Thilakkumar J   | Team Member – Project Development & Integration |
| Selva Maruthi P | Team Member – Testing & Documentation           |

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### 2. Project Overview

#### Purpose

The project focuses on importing, transforming, and validating employee data received in spreadsheet format. The solution uses a structured workflow to load employee data, stage it, map source fields to target fields, transform the data, and verify the resulting employee records.

#### Features

- Import employee data from Excel spreadsheets.
- Load spreadsheet data into an Import Set.
- Store source data in a staging table.
- Configure source-to-target field mappings.

- Transform employee data into the target table.
  - Validate required fields and data consistency.
  - Identify transformation errors.
  - Handle duplicate employee records.
  - Verify transformed employee records.
  - Reuse the Import Set and Transform Map workflow for future imports.
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### **3. Architecture**

#### **Frontend**

The project primarily uses the ServiceNow platform interface for performing the employee-data import workflow.

The interface allows administrators to:

- Upload the Excel spreadsheet.
- View imported/staged data.
- Configure Transform Maps.
- Run transformations.
- Review transformation results.
- Verify employee target records.

#### **Backend**

The ServiceNow platform manages the server-side processing of:

- Import Set processing
- Transform Map execution
- Field mapping

- Data transformation
- Record creation/update
- Error handling

## **Database**

The project uses ServiceNow tables for storing and processing employee data:

Excel Spreadsheet → Import Set → Import Set Table → Transform Map → Employee Target Table

The Import Set Table acts as the staging layer before records are transformed into the target employee table.

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## **4. Setup Instructions**

### **Prerequisites**

- ServiceNow instance
- Valid ServiceNow user account
- Administrator/import permissions
- Employee data in .xlsx format
- Web browser

### **Installation / Setup**

1. Log in to the ServiceNow instance.
2. Prepare the employee Excel spreadsheet.
3. Open System Import Sets.
4. Select Load Data.

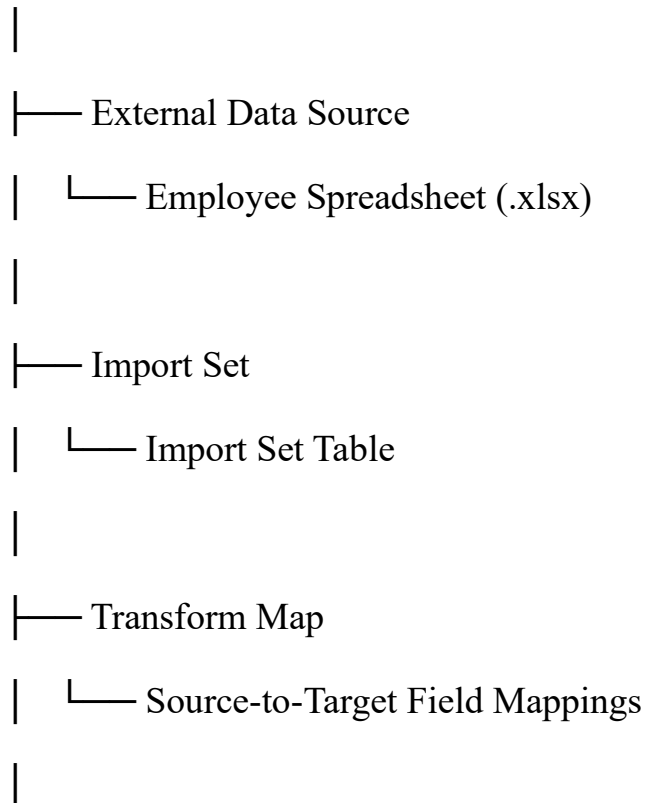
5. Upload the .xlsx file.
  6. Create or select the Import Set table.
  7. Review the imported source data.
  8. Create a Transform Map.
  9. Map source fields to target employee fields.
  10. Run the transformation.
  11. Verify the transformed employee records.
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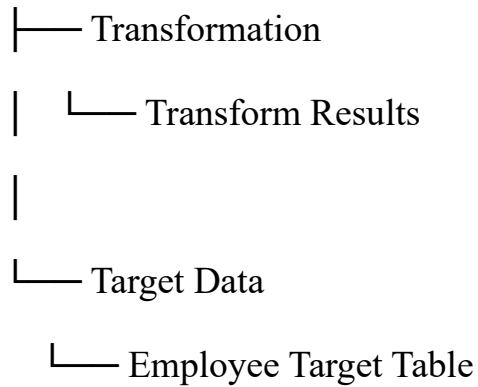
## 5. Folder Structure

Since this project is implemented using the ServiceNow platform, a traditional MERN client/server folder structure is not required.

The logical project structure is:

Import Data using Transform Maps





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## 6. Running the Application

The project runs inside the ServiceNow instance rather than through npm start.

### Steps

1. Open the ServiceNow instance.
2. Navigate to System Import Sets.
3. Select Load Data.
4. Upload the employee spreadsheet.
5. Open the Import Set table.
6. Configure/open the Transform Map.
7. Run the transformation.
8. Verify the target employee records.

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## 7. API Documentation

No custom REST API is required for the basic implementation of this project.

The main operations are performed through ServiceNow's built-in Import Set and Transform Map functionality:

| Operation      | ServiceNow Component             | Purpose                             |
|----------------|----------------------------------|-------------------------------------|
| Data Upload    | Import Set                       | Loads spreadsheet data              |
| Data Staging   | Import Set Table                 | Temporarily stores source records   |
| Field Mapping  | Transform Map                    | Maps source fields to target fields |
| Transformation | Transform Map                    | Creates/updates target records      |
| Verification   | Target Table / Transform Results | Confirms successful processing      |

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## 8. Authentication

Access to the project is controlled through ServiceNow authentication and role-based permissions.

Authorized users can:

- Upload employee spreadsheets.
- Access Import Sets.
- Configure Transform Maps.
- Run transformations.
- View target employee records.

Unauthorized users should not have access to import configuration or employee target records.

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## **9. User Interface**

The project interface includes:

- ServiceNow Load Data page
- Import Set staging table
- Transform Map configuration page
- Field mapping interface
- Transform execution/results page
- Employee target table

Screenshots to include

1. Excel file uploaded into Import Set.
2. Import Set staging records.
3. Transform Map field mappings.
4. Transformation results.
5. Final employee target records.

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## **10. Testing**

Testing covers the complete employee-data import workflow.

Functional Testing

- Excel file upload validation
- Import Set data loading
- Required-field validation

- Transform Map field mapping
- Data transformation
- Transformation error handling
- Duplicate record handling
- Target-record verification

#### Performance Testing

- Import processing time
- Bulk employee-data processing
- Large-batch transformation

#### Test Result

Total Test Cases: 19

Passed: 19

Failed: 0

Not Tested: 0

Overall Pass Percentage: 100%

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## 11. Screenshots or Demo

The project demonstration should show the complete workflow:

Employee Excel Spreadsheet

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ServiceNow Import Set

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Import Set Staging Table



Transform Map



Transformation Results



Employee Target Records

Screenshots of each stage can be included in the final project report.

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## **12. Known Issues**

- Spreadsheet column names must match the expected source fields.
- Missing or invalid employee values may require correction before transformation.
- Changes to the spreadsheet structure may require updates to the Transform Map.
- Incorrect field mappings can result in incorrect target records.
- Duplicate employee records need appropriate matching/Coalesce configuration.

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## **13. Future Enhancements**

- Add automated validation before transformation.
- Add automated duplicate detection.
- Provide detailed import-error reports.
- Support additional spreadsheet formats.

- Add automated notifications for failed imports.
  - Add scheduled employee-data imports.
  - Improve data-quality monitoring and reporting.
  - Add stronger validation rules for employee fields.
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### **Team Contribution Summary**

| Member          | Primary Contribution              |
|-----------------|-----------------------------------|
| Yogesh P        | Data Import & Testing             |
| Tarun R         | Data Validation & Documentation   |
| Ranjith R       | Transform Map & Field Mapping     |
| THILAKKUMAR J   | Project Development & Integration |
| Selva Maruthi P | Testing & Documentation           |

### **Project Flow**

Excel Employee Data → Import Set → Staging Table → Transform Map  
→ Transformation → Validation → Employee Target Records